In [2]:
```
import pandas as pd
```

In [3]:
```
data = pd.read_csv('C:\\Rprograms\\adult.csv',na_values = '?')
data
```

Out[3]:

	age	workclass	fnlwgt	education	educational-num	maritalstatus	occupation	relationship	race	gender	capitalgain	capitalloss	hour_sper_week	nativecountry	income
0	25	Private	226802	11th	7	Never-married	Machine-op-inspct	Own-child	Black	Male	0	0	40	United-States	<=50K
1	38	Private	89814	HS-grad	9	Married-civ-spouse	Farming-fishing	Husband	White	Male	0	0	50	United-States	<=50K
2	28	Local-gov	336951	Assoc-acdm	12	Married-civ-spouse	Protective-serv	Husband	White	Male	0	0	40	United-States	>50K
3	44	Private	160323	Some-college	10	Married-civ-spouse	Machine-op-inspct	Husband	Black	Male	7688	0	40	United-States	>50K
4	18	NaN	103497	Some-college	10	Never-married	NaN	Own-child	White	Female	0	0	30	United-States	<=50K
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
48837	27	Private	257302	Assoc-acdm	12	Married-civ-spouse	Tech-support	Wife	White	Female	0	0	38	United-States	<=50K
48838	40	Private	154374	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	0	0	40	United-States	>50K
48839	58	Private	151910	HS-grad	9	Widowed	Adm-clerical	Unmarried	White	Female	0	0	40	United-States	<=50K
48840	22	Private	201490	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0	20	United-States	<=50K
48841	52	Self-emp-inc	287927	HS-grad	9	Married-civ-spouse	Exec-managerial	Wife	White	Female	15024	0	40	United-States	>50K

48842 rows × 15 columns

In [4]:
```
data.maritalstatus.value_counts()
```

Out[4]:
```
Married-civ-spouse       22379
Never-married            16117
Divorced                  6633
Separated                 1530
Widowed                   1518
Married-spouse-absent      628
Married-AF-spouse           37
Name: maritalstatus, dtype: int64
```

In [5]:
```
data.groupby(["maritalstatus","gender"]).gender.count()
```

Out[5]:
```
maritalstatus          gender
Divorced               Female     4001
                       Male       2632
Married-AF-spouse      Female       25
                       Male         12
Married-civ-spouse     Female     2480
                       Male      19899
Married-spouse-absent  Female      304
                       Male        324
Never-married          Female     7218
                       Male       8899
Separated              Female      931
                       Male        599
Widowed                Female     1233
                       Male        285
Name: gender, dtype: int64
```

In [6]:
```
data.groupby(["workclass",'race','gender']).workclass.count()
```

Out[6]:
```
workclass    race                gender
?            Amer-Indian-Eskimo  Female      19
                                 Male        16
             Asian-Pac-Islander  Female      47
                                 Male        49
             Black               Female     181
                                            ...
State-gov    White               Male      1038
Without-pay  Asian-Pac-Islander  Male         1
             Black               Male         1
             White               Female       7
                                 Male        12
Name: workclass, Length: 77, dtype: int64
```

In [11]:
```
data.dropna()
```

Out[11]:

	age	workclass	fnlwgt	education	educational-num	maritalstatus	occupation	relationship	race	gender	capitalgain	capitalloss	hour_sper_week	nativecountry	income
0	25	Private	226802	11th	7	Never-married	Machine-op-inspct	Own-child	Black	Male	0	0	40	United-States	<=50K
1	38	Private	89814	HS-grad	9	Married-civ-spouse	Farming-fishing	Husband	White	Male	0	0	50	United-States	<=50K
2	28	Local-gov	336951	Assoc-acdm	12	Married-civ-spouse	Protective-serv	Husband	White	Male	0	0	40	United-States	>50K
3	44	Private	160323	Some-college	10	Married-civ-spouse	Machine-op-inspct	Husband	Black	Male	7688	0	40	United-States	>50K
5	34	Private	198693	10th	6	Never-married	Other-service	Not-in-family	White	Male	0	0	30	United-States	<=50K
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
48837	27	Private	257302	Assoc-acdm	12	Married-civ-spouse	Tech-support	Wife	White	Female	0	0	38	United-States	<=50K
48838	40	Private	154374	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	0	0	40	United-States	>50K
48839	58	Private	151910	HS-grad	9	Widowed	Adm-clerical	Unmarried	White	Female	0	0	40	United-States	<=50K
48840	22	Private	201490	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0	20	United-States	<=50K
48841	52	Self-emp-inc	287927	HS-grad	9	Married-civ-spouse	Exec-managerial	Wife	White	Female	15024	0	40	United-States	>50K

45222 rows × 15 columns

In [13]:
```
data.workclass.unique()
```

Out[13]:
```
array(['Private', 'Local-gov', nan, 'Self-emp-not-inc', 'Federal-gov',
       'State-gov', 'Self-emp-inc', 'Without-pay', 'Never-worked'],
      dtype=object)
```

In [15]:
```
data.workclass.value_counts()
```

Out[15]:
```
Private             33906
Self-emp-not-inc     3862
Local-gov            3136
State-gov            1981
Self-emp-inc         1695
Federal-gov          1432
Without-pay            21
Never-worked           10
Name: workclass, dtype: int64
```

In [7]:
```
data[data['income'] == '>50K'].groupby('race').race.count()
```

Out[7]:
```
race
Amer-Indian-Eskimo       55
Asian-Pac-Islander      409
Black                   566
Other                    50
White                 10607
Name: race, dtype: int64
```

In [ ]: